Identity and Access Management

Identity and access management, often abbreviated as IAM, refers to the framework of policies and technologies used to ensure that the right individuals have appropriate access to technology resources within an organization. At its core, IAM systems manage digital identities, verifying that users are who they claim to be through authentication processes, and controlling what actions those users are permitted to perform through authorization mechanisms. Multi-factor authentication has become a widely adopted security practice, requiring users to provide two or more forms of verification, such as a password combined with a one-time code sent to a mobile device, significantly reducing the risk of unauthorized access even if a password is compromised. The principle of least privilege is a foundational concept in access management, dictating that users should only be granted the minimum level of access necessary to perform their job functions, thereby limiting the potential damage that could result from a compromised account. Role-based access control simplifies the management of permissions by assigning access rights based on a user's role within an organization rather than configuring permissions individually for each person. Single sign-on solutions allow users to authenticate once and gain access to multiple related systems without needing to log in separately to each one, improving both security and user convenience. Regular access reviews, where organizations periodically audit who has access to which resources, help identify and revoke unnecessary permissions that may accumulate over time as employees change roles or leave the organization. Effective identity and access management is considered one of the most important defenses against both external attacks and insider threats, since compromised or overly broad access credentials remain among the leading causes of security incidents.